Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**Aim: Use appropriate smoothing and sharpening techniques to enhance the clarity of the images..**	
OPE: 9	**Date: 25/10/2026**	**Enrollment No: 92400133079**

Aim: Use appropriate smoothing and sharpening techniques to enhance the clarity of the images.

Steps:

- Libraries:- cv2(image loading), matplotlib(image plotting), numpy(mathematical calculations).
- We will load 5 given images using cv2.
- Storing all loaded image in images array.
- Defining kernel size and sigma for further usage.
- For smoothening the image we will use gaussian kernel where kernel size and sigma are taken in consideration.
- Making a user defined function Smoothening where image is processed with gaussian kernel process.
- For sharpening the image sharpening_kernel is defined within a 3X3 matrix in array.
- Making a user defined function Sharpening where image is processed with sharpening_kernel process.
- Making a for loop for plotting 5 different original with their other 4 modified images.
- Defining og(original), Si(smoothening og Image) and Shi(Sharpening og Image).
- Creating plots to display Images OG, SI & Shi.
- Creating 1 other plot where fusion of sharpened image is done with smooth image operation is SI+Shi.

Image-1:



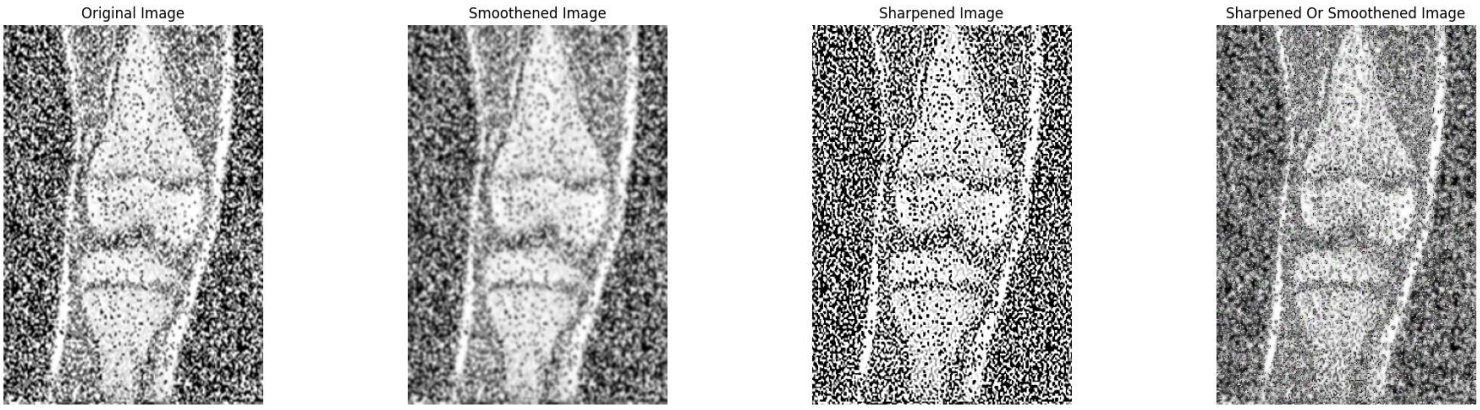
Observation:

1. The original image contains a few white spots and appears slightly blurred.
2. The smoothed image has fewer white spots and looks clearer than the original image.
3. The sharpened image highlights the white spots more prominently due to the sharpening filter applied using an array.
4. The image that is sharpened first and then smoothed appears highly noisy, with the noise becoming more noticeable.

Conclusion:

The original image was smoothed to reduce noise and improve its clarity, giving the best overall result. The sharpened image made the details more prominent but also highlighted the noise. However, applying sharpening first and smoothening afterward significantly reduced the image quality, resulting in poor clarity and excessive noise.

Image-2:



Observation:

1. The original image contains several white spots and appears slightly blurred.
2. The smoothed image has fewer white spots and looks clearer compared to the original.
3. The sharpened image makes the white spots more prominent because the sharpening filter enhances image details.
4. The image that is sharpened first and then smoothed becomes excessively noisy, resulting in a noticeable loss of image quality.

Conclusion:

The original image was smoothened to reduce white spots and improve overall clarity, giving the best visual result. The sharpened image makes the details more visible but also increases the noise. Applying sharpening first and then smoothening improves the visibility of certain details, but it introduces excessive noise and reduces the overall image quality.

Image-3:



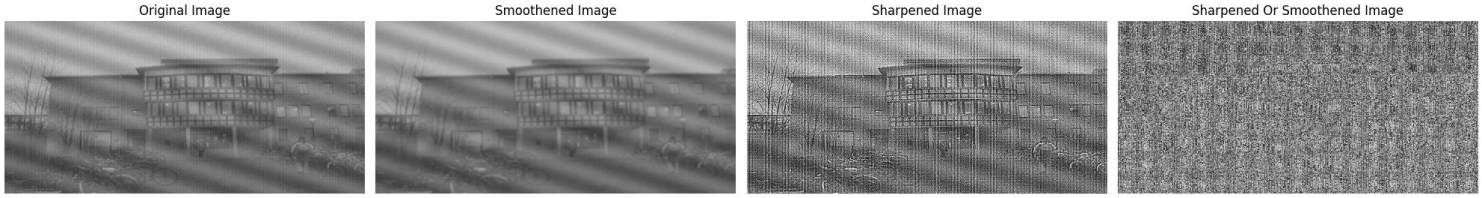
Observation:

1. The original image has some white spots and appears slightly unclear in certain areas.
2. The smoothened image shows a reduction in white spots and provides better clarity than the original.
3. The sharpened image makes the white spots more noticeable because the sharpening array enhances the image details.
4. The image processed through sharpening first and smoothening afterward becomes very noisy, which affects its overall appearance and quality.

Conclusion:

I conclude it by saying that original image was converted for smoothening which proves best clarity for image then sharpened image is perfectly visible but noisy and mixing original->sharpening->smoothening prove better image clarity but adds too much noise to it.

Image-4:



Observation:

1. It is original image some what having white dots and some blurry
2. It is smoothened image which reduced white dots and make some more clear.
3. It is sharpened image and white dots are highlighted more as it uses array for sharpening.
4. This is first sharpened then smoothened image which totally becomes noisy.

Conclusion:

I conclude it by saying that original image was converted for smoothening which proves moderate clarity for image then sharpened image is moderately visible but noisy and mixing original->sharpening->smoothening has destroyed image clarity and is too much noisy .

Image-5:



Observation:

1. It is original image some what having white dots and some blurry
2. It is smoothened image which reduced white dots and make some more clear.
3. It is sharpened image and white dots are highlighted more as it uses array for sharpening.
4. This is first sharpened then smoothened image which totally becomes noisy.

Conclusion:

I conclude it by saying that original image was converted for smoothening which proves same image as original then sharpened image is moderately visible but noisy and mixing original->sharpening->smoothening prove blurry and distorted image.

Github : https://github.com/yugmakwana99/DSIP.git